

Module 2

1. E-Commerce Platform Search Function

Code:

```
class Product {  
    int productId;  
    String productName;  
    String category;  
  
    Product(int id, String name, String category) {  
        this.productId = id;  
        this.productName = name;  
        this.category = category;  
    }  
}  
  
public class ECommerceSearch {  
    // Linear Search  
    static Product linearSearch(Product[] products, int id) {  
        for (Product p : products) {  
            if (p.productId == id) {  
                return p;  
            }  
        }  
        return null;  
    }  
  
    // Binary Search  
    static Product binarySearch(Product[] products, int id) {
```

```
int low = 0;
int high = products.length - 1;
while (low <= high) {
    int mid = (low + high) / 2;
    if (products[mid].productId == id)
        return products[mid];
    else if (products[mid].productId < id)
        low = mid + 1;
    else
        high = mid - 1;
}
return null;
}
```

```
public static void main(String[] args) {
    // Sorted array by productId
    Product[] products = {
        new Product(101, "Laptop", "Electronics"),
        new Product(102, "Phone", "Electronics"),
        new Product(103, "Shoes", "Fashion"),
        new Product(104, "Watch", "Accessories"),
        new Product(105, "Book", "Education")
    };
    int searchId = 103;
    Product result1 = linearSearch(products, searchId);
    if (result1 != null)
        System.out.println("Linear Search: " + result1.productName);
    else
        System.out.println("Product not found.");
}
```

```

        Product result2 = binarySearch(products, searchId);

        if (result2 != null)

            System.out.println("Binary Search: " + result2.productName);

        else

            System.out.println("Product not found.");

    }

}

```

Output:

```

[Running] cd "d:\Documents\Internships and
Hackathons\Cognizant Java Digital Nurture\Coding\Module 2\" &&
javac ECommerceSearch.java && java ECommerceSearch
Linear Search: Shoes
Binary Search: Shoes

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```

2. Financial Forecasting

Code:

```

public class FinancialForecasting {

    // Recursive method to calculate future value

    static double futureValue(double presentValue, double growthRate, int years) {

        // Base case

        if (years == 0) {

            return presentValue;

        }

        // Recursive case

        return (1 + growthRate) * futureValue(presentValue, growthRate, years - 1);

    }

    public static void main(String[] args) {

```

```
double presentValue = 10000; // Initial investment
double growthRate = 0.10;    // 10% annual growth
int years = 5;
double result = futureValue(presentValue, growthRate, years);
System.out.println("Present Value: " + presentValue);
System.out.println("Growth Rate: " + (growthRate * 100) + "%");
System.out.println("Years: " + years);
System.out.println("Future Value: " + result);
}
}
```

Output:

```
[Running] cd "d:\Documents\Internships and
Hackathons\Cognizant Java Digital Nurture\Coding\Module 2\" &&
javac FinancialForecasting.java && java FinancialForecasting
Present Value: 10000.0
Growth Rate: 10.0%
Years: 5
Future Value: 16105.100000000008

[Done] exited with code=0 in 1.95 seconds
```